Gemini, T., Anil, R., Borgeaud, S., Wu, Y., Alayrac, J.-B., Yu, J., Soricut, R., Schalkwyk, J., Dai, A. M., Hauth, A., et al. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023.

Geng, X., Gudibande, A., Liu, H., Wallace, E., Abbeel, P., Levine, S., and Song, D. Koala: A dialogue model for academic research. Blog post, April 2023. URL https://bair.berkeley.edu/blog/2023/04/03/koala/.

Grootendorst, M. Bertopic: Neural topic modeling with a class-based tf-idf procedure. *arXiv preprint arXiv:2203.05794*, 2022.

Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., and Steinhardt, J. Measuring massive multitask language understanding. In *International Conference on Learning Representations*, 2020.

Howard, S. R., Ramdas, A., McAuliffe, J., and Sekhon, J. Time-uniform chernoff bounds via nonnegative supermartingales. 2020.

Huang, Y., Lin, Z., Liu, X., Gong, Y., Lu, S., Lei, F., Liang, Y., Shen, Y., Lin, C., Duan, N., et al. Competition-level problems are effective llm evaluators. *arXiv preprint arXiv:2312.02143*, 2023.

Huber, P. J. et al. The behavior of maximum likelihood estimates under nonstandard conditions. In *Proceedings of the fifth Berkeley symposium on mathematical statistics and probability*, volume 1, pp. 221–233. Berkeley, CA: University of California Press, 1967.

Hunter, D. R. MM algorithms for generalized Bradley-Terry models. *The Annals of Statistics*, 32(1):384 – 406, 2004. doi: 10.1214/aos/1079120141. URL https://doi.org/10.1214/aos/1079120141.

Karimi, M. R., Gürel, N. M., Karlaš, B., Rausch, J., Zhang, C., and Krause, A. Online active model selection for pre-trained classifiers. In *International Conference on Artificial Intelligence and Statistics*, pp. 307–315. PMLR, 2021.

Karpinska, M., Akoury, N., and Iyyer, M. The perils of using Mechanical Turk to evaluate open-ended text generation. In Moens, M.-F., Huang, X., Specia, L., and Yih, S. W.-t. (eds.), *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pp. 1265–1285, Online and Punta Cana, Dominican Republic, November 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.emnlp-main.97. URL https://aclanthology.org/2021.emnlp-main.97.

Kiela, D., Bartolo, M., Nie, Y., Kaushik, D., Geiger, A., Wu, Z., Vidgen, B., Prasad, G., Singh, A., Ringshia, P., et al. Dynabench: Rethinking benchmarking in nlp. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 4110–4124, 2021.

Köpf, A., Kilcher, Y., von Rütte, D., Anagnostidis, S., Tam, Z.-R., Stevens, K., Barhoum, A., Duc, N. M., Stanley, O., Nagyfi, R., et al. Openassistant conversations–democratizing large language model alignment. *arXiv preprint arXiv:2304.07327*, 2023.

Langley, P. Crafting papers on machine learning. In Langley, P. (ed.), *Proceedings of the 17th International Conference on Machine Learning (ICML 2000)*, pp. 1207–1216, Stanford, CA, 2000. Morgan Kaufmann.

Li, X., Zhang, T., Dubois, Y., Taori, R., Gulrajani, I., Guestrin, C., Liang, P., and Hashimoto, T. B. Alpacaeval: An automatic evaluator of instruction-following models. https://github.com/tatsu-lab/alpaca_eval, 2023.

Li, Y., Choi, D., Chung, J., Kushman, N., Schrittwieser, J., Leblond, R., Eccles, T., Keeling, J., Gimeno, F., Dal Lago, A., et al. Competition-level code generation with alphacode. *Science*, 378(6624):1092–1097, 2022.

Liang, P., Bommasani, R., Lee, T., Tsipras, D., Soylu, D., Yasunaga, M., Zhang, Y., Narayanan, D., Wu, Y., Kumar, A., et al. Holistic evaluation of language models. *arXiv preprint arXiv:2211.09110*, 2022.

Lin, Z., Wang, Z., Tong, Y., Wang, Y., Guo, Y., Wang, Y., and Shang, J. ToxicChat: Unveiling hidden challenges of toxicity detection in real-world user-AI conversation. In Bouamor, H., Pino, J., and Bali, K. (eds.), *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 4694–4702, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.311. URL https://aclanthology.org/2023.findings-emnlp.311.

Liu, T.-Y. et al. Learning to rank for information retrieval. *Foundations and Trends® in Information Retrieval*, 3(3): 225–331, 2009.

McInnes, L., Healy, J., and Melville, J. Umap: Uniform manifold approximation and projection for dimension reduction, 2020.

OpenAI. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.

Oren, Y., Meister, N., Chatterji, N., Ladhak, F., and Hashimoto, T. B. Proving test set contamination in black box language models. *arXiv preprint arXiv:2310.17623*, 2023.

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C. L., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., Schulman, J., Hilton, J., Kelton, F., Miller, L., Simens, M., Askell, A., Welinder, P., Christiano, P., Leike, J., and Lowe, R. Training language models to follow instructions with human feedback, 2022.

Ramdas, A., Grünwald, P., Vovk, V., and Shafer, G. Game-theoretic statistics and safe anytime-valid inference. *Statistical Science*, 38(4):576–601, 2023.

Rao, P. V. and Kupper, L. L. Ties in paired-comparison experiments: A generalization of the bradley-terry model. *Journal of the American Statistical Association*, 62(317): 194–204, 1967. doi: 10.1080/01621459.1967.10482901.

Srivastava, A., Rastogi, A., Rao, A., Shoeb, A. A. M., Abid, A., Fisch, A., Brown, A. R., Santoro, A., Gupta, A., Garriga-Alonso, A., et al. Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. *Transactions on Machine Learning Research*, 2023.

Szörényi, B., Busa-Fekete, R., Paul, A., and Hüllermeier, E. Online rank elicitation for plackett-luce: A dueling bandits approach. In Cortes, C., Lawrence, N., Lee, D., Sugiyama, M., and Garnett, R. (eds.), *Advances in Neural Information Processing Systems*, volume 28. Curran Associates, Inc., 2015. URL https://proceedings.neurips.cc/paper_files/paper/2015/file/7eacb532570ff6858afd2723755ff790-Paper.pdf.

Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., Bashlykov, N., Batra, S., Bhargava, P., Bhosale, S., et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.

Vovk, V. and Wang, R. E-values: Calibration, combination and applications. *The Annals of Statistics*, 49(3):1736–1754, 2021.

Wang, Y., Kordi, Y., Mishra, S., Liu, A., Smith, N. A., Khashabi, D., and Hajishirzi, H. Self-instruct: Aligning language models with self-generated instructions. In Rogers, A., Boyd-Graber, J., and Okazaki, N. (eds.), *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 13484–13508, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-long.754. URL https://aclanthology.org/2023.acl-long.754.

Waudby-Smith, I. and Ramdas, A. Estimating means of bounded random variables by betting. *arXiv preprint arXiv:2010.09686*, 2020.

White, H. Maximum likelihood estimation of misspecified models. *Econometrica: Journal of the econometric society*, pp. 1–25, 1982.

Yang, S., Chiang, W.-L., Zheng, L., Gonzalez, J. E., and Stoica, I. Rethinking benchmark and contamination for language models with rephrased samples. *arXiv preprint arXiv:2311.04850*, 2023.

Zellers, R., Holtzman, A., Bisk, Y., Farhadi, A., and Choi, Y. Hellaswag: Can a machine really finish your sentence? In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 4791–4800, 2019.

Zheng, L., Chiang, W.-L., Sheng, Y., Li, T., Zhuang, S., Wu, Z., Zhuang, Y., Li, Z., Lin, Z., Xing, E. P., Gonzalez, J. E., Stoica, I., and Zhang, H. Lmsys-chat-1m: A large-scale real-world llm conversation dataset, 2023a.

Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E., Zhang, H., Gonzalez, J. E., and Stoica, I. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2023b. URL https://openreview.net/forum?id=uccHPGDlao.

Zhong, W., Cui, R., Guo, Y., Liang, Y., Lu, S., Wang, Y., Saied, A., Chen, W., and Duan, N. Agieval: A human-centric benchmark for evaluating foundation models. *arXiv preprint arXiv:2304.06364*, 2023.

Zhu, B., Frick, E., Wu, T., Zhu, H., and Jiao, J. Starling-7b: Improving llm helpfulness & harmlessness with rlaif, November 2023.

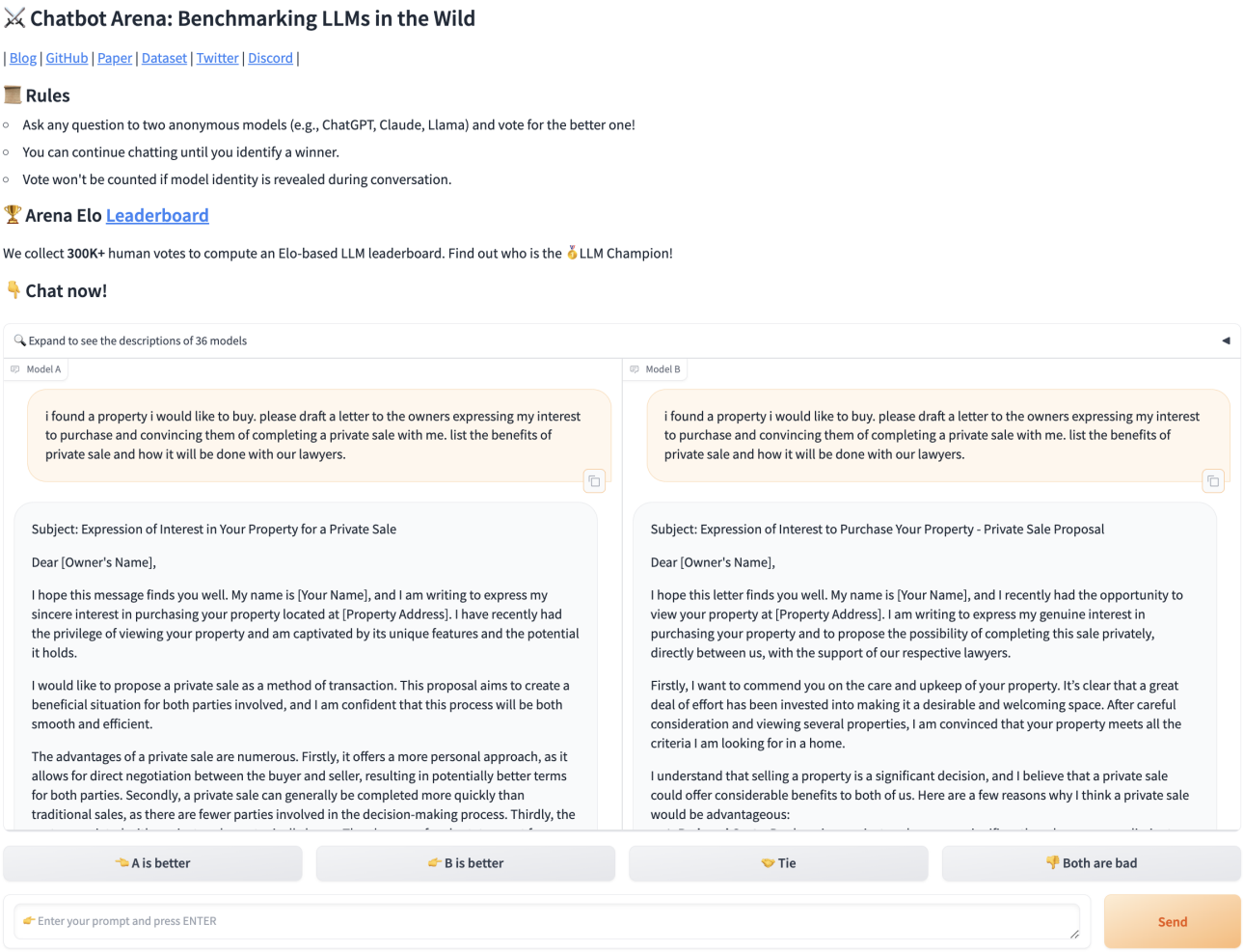

Figure 8. Screenshot of Chatbot Arena.

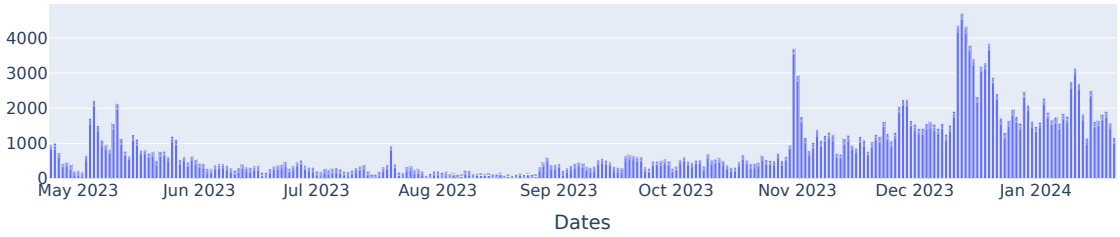

Figure 9. The number of votes over time